

Department of Electrical and Computer Engineering
COMSATS University Islamabad, Attock Campus

Lab Rubrics Form	
Lab Name	User-Defined Functions in C Programming
Lab Number	03
Submitted by:	
Student Names	Registration #
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Rubrics		Marks		
		1	2	3
PLO-5 Modern Tools Usage	Ability to execute the experiment utilizing hardware, software tools, or a combination of both.			
PLO-10 Communication Skills	Present lab activities effectively through detailed lab reports, oral examinations, and engaging presentations.			

Lab 03: User-Defined Functions in C Programming

Objective

To understand the concept of user-defined functions in C programming, including passing arguments, returning values, performing calculations through functions, and improving program modularity.

Software Requirements

- Code::Blocks IDE / Dev-C++
- GCC Compiler.
- Windows Operating System.

Experiments Performed

In-Lab Tasks

1. Perform addition, subtraction, multiplication, and division using user-defined functions.
2. Calculate the area of a circle, rectangle, and square using separate functions.

Post-Lab Tasks

1. Swap two numbers inside a function without using a third variable.
2. Pass a number to a function to calculate and return its square.
3. Input a four-digit number and calculate the sum of its first and last digits using a function.

Learning Outcomes

After completing this lab, the student will be able to:

- Understand the concept of user-defined functions.
- Pass arguments to functions and receive returned values.
- Divide a program into reusable modules.
- Implement mathematical calculations using functions.
- Apply functions to solve simple programming problems.

In lab Task:

Task 1:

Write a program to print sum, subtract, division, multiplications using functions?

Code:

```
#include <stdio.h>

#include <stdlib.h>

int main()
{
    int x, y;

    int Sum, Subtract, Multiply, Divide;

    printf("Enter the value of x:\n",x);
    scanf("%d", &x);

    printf("Enter the value of y:\n",y);
    scanf("%d", &y);

    Sum= calsum(x,y);
    Subtract= calsubtract(x,y);
    Multiply= calmultiply(x,y);
    Divide= caldivide(x,y);

    printf("\t The sum = %d\n", Sum);
    printf("\t The subtraction = %d\n", Subtract);
    printf("\t The multiplication = %d\n", Multiply);
    printf("\t The division= %d\n", Divide);

    return 0;
}

calsum(int p, int q)
{
    int r;
```

```

    r=p+q;
    return (r);
}
calsubtract(int p, int q)
{
    int r;
    r=p-q;
    return (r);
}
calmultiply(int p, int q)
{
    int r;
    r=p*q;
    return (r);
}
caldivide(int p, int q)
{
    int r;
    r=p/q;
    return (r);
}

```

Output:

```

C:\Users\Hamza Computer\Documents\PF\lab1\lab2\bin\Debug\lab2.exe
Enter the value of x:
9
Enter the value of y:
6
    The sum = 15
    The subtraction = 3
    The multiplication = 54
    The division= 1

Process returned 0 (0x0)   execution time : 4.195 s
Press any key to continue.

```

Task 2:

Write a program to print area of Circle, Rectangle, Square using functions.

Code:

```
#include <stdio.h>

#include <stdlib.h>

float calcircle(float r);

float calrectangle(float L, float W);

float calsquare(float L);

int main()
{
    float r, L, W, Area_of_C, Area_of_R, Area_of_S;

    printf("\t Enter the value of r:");
    scanf("%f", &r);

    printf("\t Enter the value of L:");
    scanf("%f", &L);

    printf("\t Enter the value of W:");
    scanf("%f", &W);

    Area_of_C=calcircle(r);
    Area_of_R=calrectangle(L,W);
    Area_of_S=calsquare(L);


    printf("\t Area of circle= %f\n", Area_of_C);
    printf("\t Area of rectangle= %f\n", Area_of_R);
    printf("\t Area of square= %f\n", Area_of_S);
    return (0);
}
```

```

}

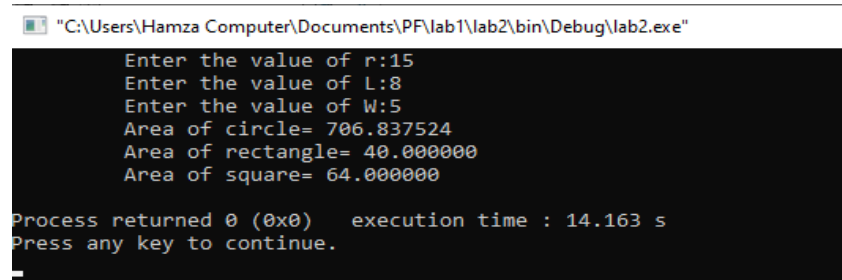
float calcircle(float r)
{
    float x;
    x=3.1415*r*r;
    return (x);
}

float calrectangle(float L, float W)
{
    float y;
    y=L*W;
    return (y);
}

float calsquare(float L)
{
    float z;
    z=L*L;
    return (z);
}

```

Output:



```

"C:\Users\Hamza Computer\Documents\PF\lab1\lab2\bin\Debug\lab2.exe"
Enter the value of r:15
Enter the value of L:8
Enter the value of W:5
Area of circle= 706.837524
Area of rectangle= 40.000000
Area of square= 64.000000
Process returned 0 (0x0)   execution time : 14.163 s
Press any key to continue.

```

Post Lab Task:

Task 1:

Write a program which input two values in main and send it to the function. Function swap these two values without using third variable and print in the Function?

Code:

```
#include <stdio.h>

#include <stdlib.h>

int main()
{
    int a,b;

    printf("\nEnter the value of a and b:\n");

    scanf("%d %d", &a,&b);

    /* wapping the value */

    printf("before swaping \n X=%d \nY=%d", a,b);

    return 0;
}
```

Output:

Task 2:

Write a Program which input a number and pass it by function. The function takes square of the number and return it.

Code:

```
#include <stdio.h>
```

```

#include <stdlib.h>

int main()
{
    int l, square;

    printf("Enter a number to square=");

    scanf("%d",&l);

    square=square_of(l);

    return 0;
}

int square_of(int l)
{
    int s;

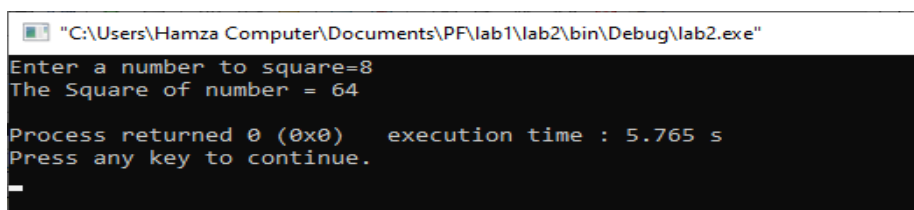
    s=l*l;

    printf("The Square of number = %d\n",s);

    return 0;
}

```

Output:



```

"C:\Users\Hamza Computer\Documents\PF\lab1\lab2\bin\Debug\lab2.exe"
Enter a number to square=8
The Square of number = 64

Process returned 0 (0x0)   execution time : 5.765 s
Press any key to continue.
_

```

Task 3:

Write a program which input a 4-digit number and send it to the function. Function should calculate the sum of 1st and last digit of this number and return it to main.

Code:

```

#include <stdio.h>

#include <stdlib.h>

```



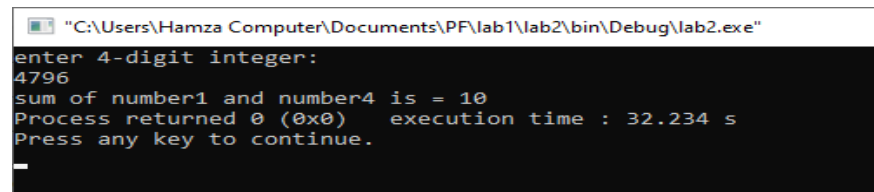
```

int main()
{
    int number1,number4,num,sum;
    printf("enter 4-digit integer:\n");
    scanf("%d",&num);
    sum=calsum(number1,number4,num);
    return 0;
}

calsum(int number1,int number4,int num)
{
    int sum;
    number1=(num/1000)%10;
    number4=num%10;
    sum=number1+number4;
    printf("sum of number1 and number4 is = %d",sum);
    return 0;
}

```

Output:



```

"C:\Users\Hamza Computer\Documents\PF\lab1\lab2\bin\Debug\lab2.exe"
enter 4-digit integer:
4796
sum of number1 and number4 is = 10
Process returned 0 (0x0)   execution time : 32.234 s
Press any key to continue.

```